

$$i_k = i \left(\frac{\frac{1}{R_k}}{\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}} \right); \quad i_2 = i \left(\frac{R_1}{R_1 + R_2} \right) \quad \text{Current Division}$$

DC/AC Circuits

$$v_k = \left(\frac{R_k}{R_1 + R_2 + \dots + R_n} \right) v; \quad v_2 = \left(\frac{R_2}{R_1 + R_2} \right) v$$

Exam 1

09.18.2026

Name Ethan Moss

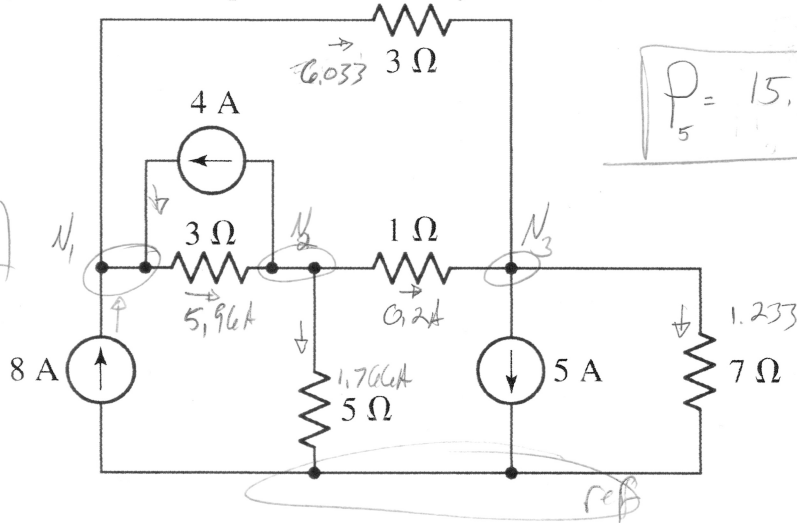
Problem 1 (25 points)

Use nodal analysis and find the power absorbed by 5 Ohm resistor

$$\begin{aligned} v_1 &= 26.733 \text{ V} \\ v_2 &= 8.833 \text{ V} \\ v_3 &= 8.633 \text{ V} \end{aligned}$$

$$I_5 = \frac{v_2}{R_5} = \frac{8.833}{5} = 1.766 \text{ A}$$

$$P_5 = (I_5)^2 \cdot R = 15.6 \text{ W}$$



$$\begin{aligned} \text{Node 1: } \frac{v_1}{3} + \frac{v_1 - v_2}{3} &= \frac{8}{1} + \frac{v_1 - v_3}{3} \Rightarrow \frac{v_1}{3} + \frac{v_1}{3} - \frac{v_2}{3} - \frac{v_3}{3} = 12 \Rightarrow 2v_1 - v_2 - v_3 = 36 \quad (1) \end{aligned}$$

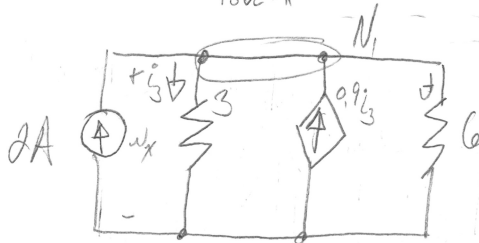
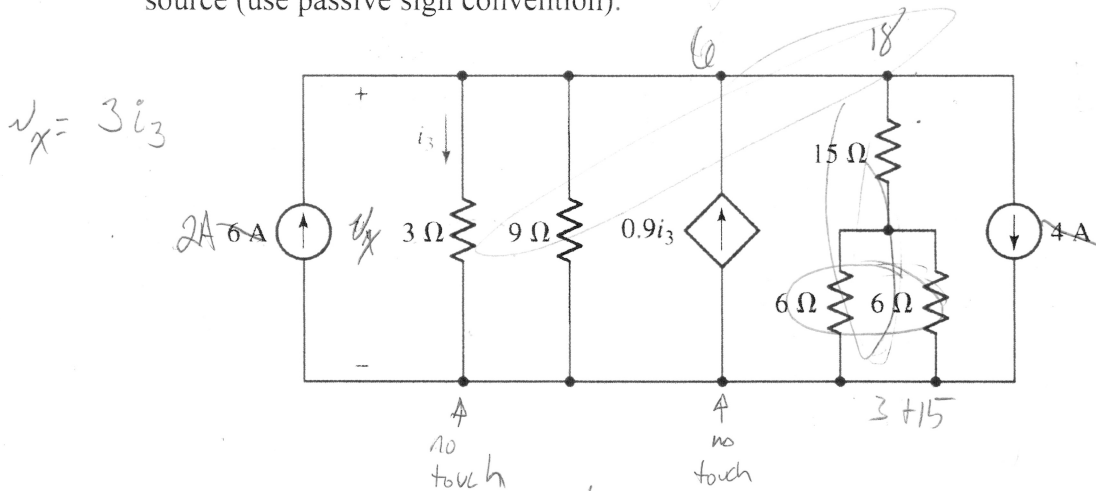
$$\begin{aligned} \text{Node 2: } \frac{v_1 - v_2}{3} &= \frac{v_2}{5} + \frac{v_2 - v_3}{1} + 4 \Rightarrow \frac{v_1}{3} - \frac{v_2}{3} - \frac{v_2}{5} - \frac{v_2 - v_3}{1} + \frac{v_3}{1} = 4 \Rightarrow 5v_1 - 23v_2 + 15v_3 = 60 \quad (2) \end{aligned}$$

$$\begin{aligned} \text{Node 3: } \frac{v_1 - v_3}{1} + \frac{v_1 - v_3}{3} &= \frac{5}{1} + \frac{v_3}{7} \Rightarrow \frac{v_1}{3} + \frac{v_2}{1} - \frac{v_3}{1} - \frac{v_3}{3} - \frac{v_3}{7} = 5 \Rightarrow 7v_1 + 21v_2 - 31v_3 = 105 \quad (3) \end{aligned}$$

Don't touch i_3 resistor

Problem 2 (25 points)

Use source/resistor combination rules and find the voltage and power on the dependent source (use passive sign convention).



Node 1

$$2 + 0.9i_3 = \frac{v_x}{6} + i_3$$

$$\frac{v_x}{6} + i_3 - 2 - 0.9i_3 = 0$$

$$\frac{3i_3}{6} + i_3 - 2 - 0.9i_3 = 0$$

$$0.5i_3 + i_3 - 0.9i_3 = 2$$

$$1.5i_3 - 0.9i_3 = 2$$

$$0.6i_3 = 2$$

$$i_3 = 3.33 \text{ A}$$

v_x = all components are in parallel
all components $v = v_x$

$$v_x = 3(3.33) \text{ A} = 9.99 \text{ A} \approx 10 \text{ V}$$

$$P_{\text{Dep}} = 10 \cdot (-3) = -30 \text{ W}$$

$$I_{\text{Dep}} = 0.9(3.33 \text{ A}) = 3 \text{ A}$$

Current enters
Positive terminal = -3A

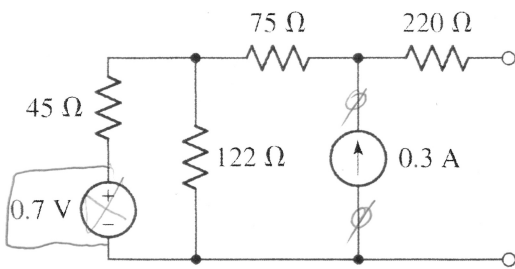
$$V_{\text{Dependent}} = 10 \text{ V}$$

$$I_{\text{Dependent}} = -3 \text{ A}$$

$$P_{\text{Dependent}} = -30 \text{ W}$$

Problem 3 (25 points)

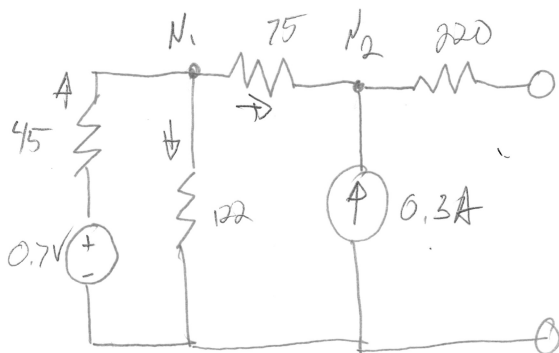
Determine Thevenin's and Norton's equivalents for the network below:



$$\frac{R_{45} \cdot R_{122}}{R_{45} + R_{122}} = 32.87 \Omega$$

$$R_{45,122} + R_{25} = 107.87 \Omega$$

$$R_{TH} = 327.87 \Omega$$



(N1)

$$\frac{0.7 - v_1}{45} = \frac{v_1}{122} + \frac{v_1 - v_2}{75}$$

$$\frac{0.7 - (v_2 - 22.5)}{45} = \frac{v_2 - 22.5}{122} + \frac{22.5 - v_2}{75}$$

$$\frac{0.7(411750)}{45} - \frac{v_2(411750)}{45} + \frac{22.5(411750)}{45} = \left(\frac{v_2}{122} - \frac{22.5}{122} - \frac{22.5}{75} \right) (411750)$$

$$\left(\frac{v_2}{122} + \frac{v_2}{45} \right) (411750) = \left(\frac{0.7}{45} + \frac{22.5}{45} + \frac{22.5}{122} + \frac{22.5}{75} \right) (411750)$$

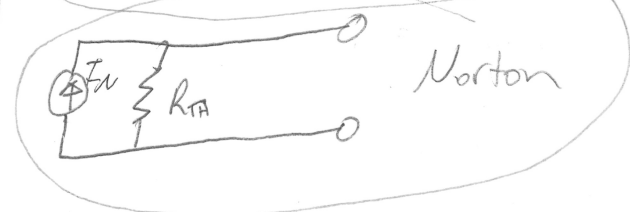
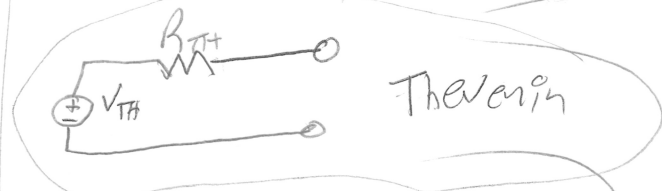
$$v_2 = \frac{(6407 + 206875 + 75937.3 + 123525)}{12525}$$

$$v_2 = 32.87 V$$

$$v_1 = 10.37 V$$

$$V_{TH} = v_2 = 32.87 V$$

$$I_N = \frac{V_{TH}}{R_{TH}} = \frac{32.87 V}{327.87 \Omega} = 0.10026 A \approx 0.1 A = I_N$$



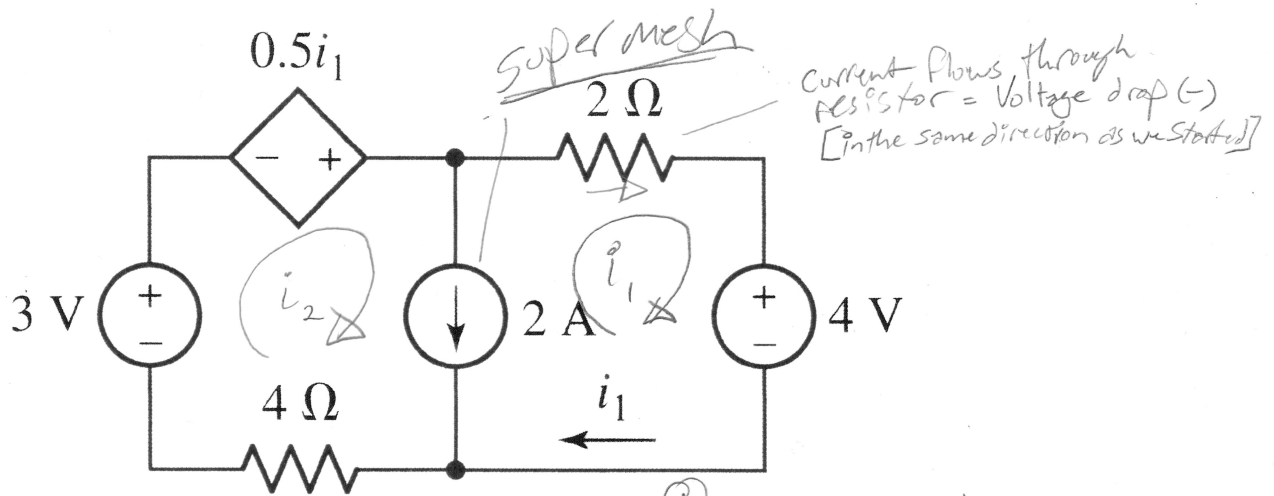
$$N_2 \left/ \frac{v_1 - v_2}{75} + 0.3 = 0 \right.$$

$$\frac{v_2}{75} = \frac{v_1}{75} + 0.3$$

$$\begin{cases} v_2 = v_1 + 22.5 \\ v_1 = v_2 - 22.5 \end{cases}$$

Problem 4 (25 points)

Use mesh analysis to find the current through $2\ \Omega$ resistor



KVL super mesh

$$\textcircled{1} \quad i_2 - i_1 = 2A$$

$$i_2 = i_1 + 2$$

$$3 + 0.5i_1 - 2(i_1) - 4(i_2) = 0$$

$$3 - 4 + 0.5i_1 - 2i_1 - 4(i_1 - 2) = 0$$

$$0.5i_1 - 2i_1 - 4i_1 - 8 = 1$$

$$-6.5i_1 + 0.5i_1 = 9$$

$$i_1 = \frac{9}{-5.5} = -1.6364\text{ A}$$

$$i_2 = -1.6364 + 2 = 0.3636\text{ A}$$

$$I_{R_2} = i_1 = -1.6\text{ A}$$